MATH317 · NUMERICAL METHODS

Chapter 3.1 Bisection Method

Midterm summary · Lecture 5

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01 What is a root?

A **Root** (or **Zero**) of a function `f` is a number `r` with `f(r) = 0`.

Some equations can be solved by hand. Example:

$f(x) = 6x^2 - 7x + 2 = (2x - 1)(3x - 2)$ roots: $\frac{1}{2}$ and $\frac{2}{3}$

Most equations have no hand solution, for example $x^3 - 2\sin x = 0$. For those we need a numerical method.

Methods of the chapter

Chapter 3 has three methods: **Bisection**, **Newton** and **Secant**. This file covers bisection, together with **False Position**.

02 The theorem behind bisection

The method rests on the **Intermediate Value Theorem**:

Theorem

If f is continuous on $[a, b]$ and $f(a) \cdot f(b) < 0$, then there is at least one root between a and b.

In words: if the function has opposite signs at the two ends, it must cross the x-axis between them.

Watch out

If $f(a) \cdot f(b) > 0$ there may or may not be a root. A sign change is sufficient, not necessary.

03 The bisection algorithm

The idea: halve the interval each time and keep the half that contains the root. It is also called the **Binary-search method**.

1

Start with `[a, b]` and compute `u = f(a)` and `v = f(b)`. We need `u · v < 0`.

2 **Compute the midpoint $c = \frac{a + b}{2}$ and `w = f(c)`.**

3 **Check the sign of `u · w`:**

Case	Meaning	Update
`u · w < 0`	The root is in `[a, c]`	`b = c`
`u · w > 0`	The root is in `[c, b]`	`a = c`
`u · w = 0`	`c` is the root	stop

4 **Go back to step 2 and repeat.**

When to stop (stopping criteria)

- When the number of iterations exceeds `n_max`.
- Or when the **Error bound** drops below `ε`.

04 A full worked example

Slide example: $f(x) = x^3 - 3x + 1$ on $[0, 1]$.

```
f(0) = 1          f(1) = 1 − 3 + 1 = −1          f(0) · f(1) < 0   →  a
root exists
```

n	a	b	c	f(c)	Error
0	0	1	0.5	−0.375	0.5
1	0	0.5	0.25	0.2656	0.25
2	0.25	0.5	0.375	−0.0723	0.125
3	0.25	0.375	0.3125	0.0930	0.0625
4	0.3125	0.375	0.34375	0.0094	0.03125

How the first two rows are read:

- $n = 0$: the midpoint is 0.5 and $f(0.5) = -0.375$. Its sign is opposite to $f(0) = 1$, so the root is in $[0, 0.5]$.
- $n = 1$: the midpoint is 0.25 and $f(0.25) = 0.2656$. Its sign matches $f(0)$, so the root is in $[0.25, 0.5]$.

The true root is 0.3472963553. The method approaches it slowly.

The Error column

In the doctor's table the Error is half the length of the current interval: it starts at $\frac{b - a}{2}$ and halves on every row.

The slide shows this table as an image, so the numbers here were computed with the same method.

05 Error bound and number of steps

After n steps (counting from $n = 0$), the distance between the midpoint c_n and the root r satisfies:

$$| r - c_n | \leq \frac{b - a}{2^{n+1}}$$

How many steps for accuracy ε?

$$\frac{b - a}{2^{n+1}} < \varepsilon$$

$$n > \frac{\log(b - a) - \log(2\varepsilon)}{\log 2}$$

Slide example

How many steps give full single precision when $a = 16$ and $b = 17$?

$$16 = (10000.0)_2 \qquad 17 = (10001.0)_2$$

The root lies between them, so the first 5 bits are known. Single precision has 24 bits, leaving 19 to determine. The inequality gives:

$$n \geq 20$$

Appeared in the midterm

A statement in midterm 472 said the bisection error after n steps is at most $\frac{b - a}{2^n}$.

The answer was False; the correct power is 2^{n+1}.

Strengths and weaknesses of bisection

Strengths	Weaknesses
Reliable: it always closes in on a root once the sign condition holds.	Slow. It does not give the exact root in a finite number of steps.

That is why it is often used to get close to a root before switching to a faster method such as Newton's.

06 False Position (Regula Falsi)

Like bisection it keeps the root bracketed between `a` and `b`. The difference: instead of the midpoint, it uses the point where the line through `(a, f(a))` and `(b, f(b))` crosses the x-axis.

$$c = \frac{a \cdot f(b) - b \cdot f(a)}{f(b) - f(a)}$$

The new interval is chosen as in bisection:

Case	New interval
`f(a) · f(c) < 0`	`[a, c]`
`f(c) · f(b) < 0`	`[c, b]`

Slide example

`f(x) = x³ − 2x − 3` on `[0, 2]`.

$$f(0) = -3 \qquad f(2) = 1 \qquad f(0) \cdot f(2) = -3 < 0$$

$$\text{Step 1:} \quad c = \frac{0 \cdot 1 - 2 \cdot (-3)}{1 - (-3)} = \frac{6}{4} = 1.5$$

$$f(1.5) = -2.625 \quad \text{(negative)} \quad \rightarrow \quad a = 1.5\,, \quad b = 2$$

$$\text{Step 2:} \quad c = \frac{1.5 \cdot 1 - 2 \cdot (-2.625)}{1 + 2.625} = \frac{6.75}{3.625} = \frac{54}{29} = 1.8621$$

$$f(1.8621) = -0.2678 \quad \text{(negative)} \quad \rightarrow \quad a = 1.8621\,, \quad b = 2$$

07 How it shows up in the midterm

The bisection question appeared in all four midterms, worth 8 to 11 marks. Its shape is fixed: part (a) shows a root exists, part (b) applies the method with `n_max = 2`.

Midterm 472 solved the doctor's way

f(x) = $-x^2 + 4x - e^x$. (a) Prove that f has exactly one real root in [0, 1]. (b) Find a root on [0, 1] using the bisection method. Terminate if iterations exceed nmax = 2 or the error bound is less than 0.5×10^{-6}.

$$\text{(a)} \quad u = f(0) = -1 \qquad v = f(1) = 0.2817$$

$$u \cdot v < 0 \quad \rightarrow \quad f \text{ has a root in } [0, 1]$$

(b) $n = 0$: $x_0 = \frac{0 + 1}{2} = 0.5$ $w = f(0.5) = 0.1013$

$u \cdot w < 0 \rightarrow b = x_0 \rightarrow [0, 0.5]$

$n = 1$: $x_1 = \frac{0 + 0.5}{2} = 0.25$ $w = f(0.25) = -0.3465$

$u \cdot w > 0 \rightarrow a = x_1 \rightarrow [0.25, 0.5]$

$n = 2$: $x_2 = \frac{0.25 + 0.5}{2} = 0.375$ $w = f(0.375) = -0.0956$

$u \cdot w > 0 \rightarrow a = x_2 \rightarrow [0.375, 0.5]$

n	x_n	$f(x_n)$	Error
0	0.5	0.1013	0.5
1	0.25	−0.3465	0.25
2	0.375	−0.0956	0.125

Notes on the answer key

The key prints `x₂ = 2.75` in the table and an error of `0.25` for the third step in the text. Both are typos; the correct values are `0.375` and `0.125`.

The doctor also notes that skipping any step loses marks. Write every line.

Past papers show the style of the question. They are not a promise about the next exam.

08 Common mistakes

Mistake	Fix
Calculator in degree mode with `sin` or `cos`	It must be in radian mode.
Writing the error bound as $\frac{b-a}{2^n}$	The correct bound is $\frac{b-a}{2^{n+1}}$.
Keeping the wrong half of the interval	Keep the half whose ends have opposite signs.
Writing only two rows when `n_max = 2`	Three rows: `n = 0, 1, 2`.
Saying bisection gives the exact root	It gives an approximation; the error halves each step.

09 Check yourself

1. `f(x) = x³ − x − 1` on `[1, 2]`. Apply bisection with `n_max = 2`.
2. `f(x) = cos x − x` on `[0, 1]` (radians). Apply bisection with `n_max = 2`.
3. How many steps on `[0, 1]` make the error smaller than `10⁻³`?

Solutions

Q1 · n	c	f(c)	Error	Next interval
0	1.5	0.875	0.5	[1, 1.5]
1	1.25	−0.2969	0.25	[1.25, 1.5]
2	1.375	0.2246	0.125	[1.25, 1.375]

Q2 · n	c	f(c)	Error	Next interval
0	0.5	0.3776	0.5	[0.5, 1]
1	0.75	−0.0183	0.25	[0.5, 0.75]
2	0.625	0.1860	0.125	[0.625, 0.75]

Q3) $\frac{1-0}{2^{n+1}} < 10^{-3} \rightarrow 2^{n+1} > 1000 \rightarrow n+1 \geq 10$

$n \geq 9$